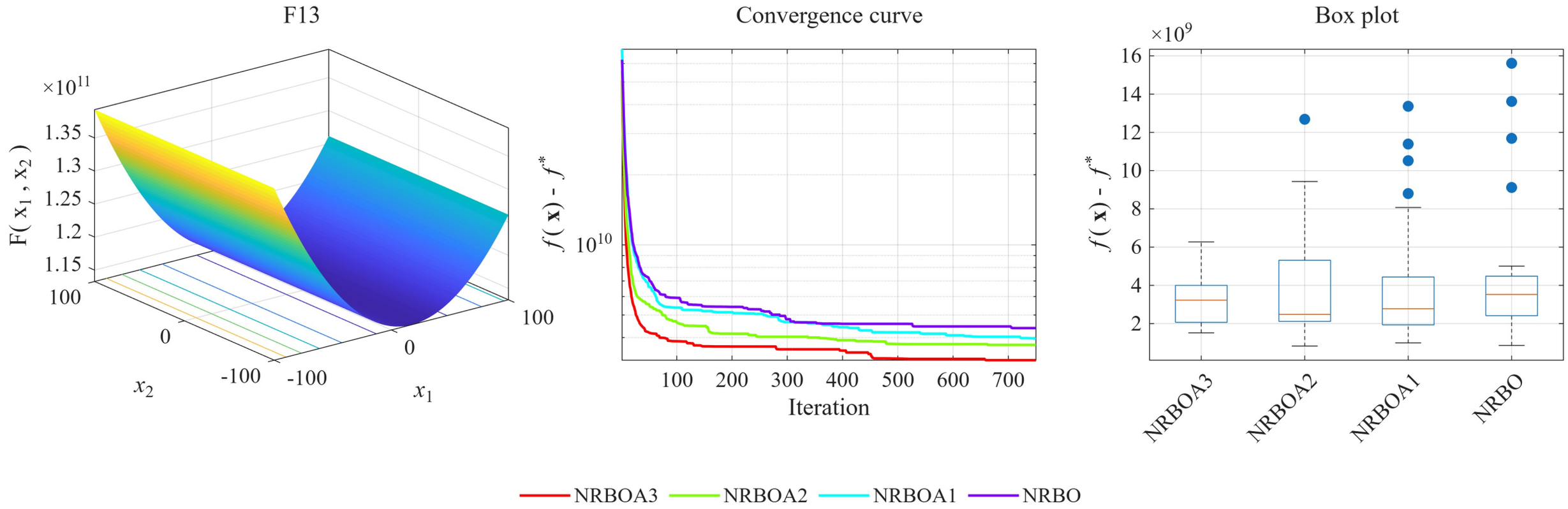

F13
×10^11
1.35
1.3
1.25
1.2
1.15
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
10^10
100
200
300
400
500
600
700
Iteration
Box plot
×10^9
16
14
12
10
8
6
4
2
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO